

Tyler Buxton

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EDUCATION

Virginia Tech

M.S. Thesis Computer Science & Applications, GPA: 4.0

May 2025

B.S. Computer Science, GPA: 3.88

May 2024

Societies: Lambda Chi Alpha, Phi Beta Kappa

RELEVANT EXPERIENCE

Full Stack Developer II, FedEx, Remote

June 2025 – Present

- Lead live maps implementation to automate weekly map installations for 50,000+ driver Android devices, eliminating ~52 hours of annual manual work and enabling international route support
- Delivered picture proof of delivery integration projected to save an estimated \$3M through reduced processing time and storage costs
- Automated multilingual localization pipeline to accurately translate app strings across English, French, Spanish, and additional languages for international markets
- Reduced Android codebase size by ~90% by introducing Lombok and HILT dependency injection, improving long-term maintainability

Graduate & Undergraduate Teaching Assistant, Blacksburg, VA

August 2022 – May 2025

- Worked as a TA for Software Design (CS 1114), Data Structures (CS 2114), and Comparative Languages (CS 3304)
- Improved leadership and communication skills by instructing labs of about 30 students

Software Engineer Intern, FedEx, Remote

June 2024 – August 2024

- Contributed to an Android mobile application used by over 50,000 drivers for pickup and delivery
- Gained experience with the agile software development process by attending daily standups, creating and completing stories, and participating in sprint planning, refinement, and retrospectives

Undergraduate Research Assistant, Blacksburg, VA

January 2023 – May 2024

- Analyzed 957 students' data using Python and Matplotlib to publish a paper in FIE 2024 conference
- Enhanced usability and added authentication workflow to React web application used to interactively teach command line and Git skills to Computer Science students
- Created backend used by instructors and students to manage classes and exercises

Software Engineer Intern, FedEx, Remote

June 2023 – August 2023

- Developed web application using Angular and TypeScript that communicated with enterprise REST APIs to reroute packages, reducing time spent rerouting packages by 50%
- Led biweekly demos for team members and presented final application to company vice presidents

PROJECTS

Git Interactive Visualization Learning Platform, M.S. Thesis, Virginia Tech

August 2023 – May 2025

- Developed a web-based platform to teach command line and Git through interactive visualizations, scaffolded exercises, and real-time feedback, supporting 5 instructors with unlimited customizable exercises
- Achieved a system usability score of 82.5 and an 8.8/10 expert recommendation score through a formal expert evaluation
- Pilot study demonstrated the platform's ability to differentiate skill levels, with experts completing exercises 3.6x faster (10 min vs. 35 min) than beginners

Empower, Personal Project

May 2025 – Present

- Built a multi-tenant scheduling platform with calendar-based weekly scheduling, time-off management, admin dashboards, and Stripe payment integration, designed to scale across unlimited tenant companies
- Developed full-stack monorepo using React, TypeScript, TanStack Router, React Query, and Tailwind CSS for the frontend
- Implemented 20+ type-safe API routes with Fastify and oRPC backed by AWS DynamoDB, secured with Clerk authentication and role-based authorization

AI Chatbot, Senior Capstone

January 2024 – May 2024

- Led 5 person team and collaborated with client to develop AI chatbot to comfort obesity and type 2 diabetes
- Integrated Llama and OpenAI LLMs to facilitate cue text generation for 100 users

SKILLS

Languages: Python, Java, TypeScript, JavaScript, HTML, CSS

Technologies: React, Android, Tailwind CSS, TanStack Router, React Query, Fastify, Zod, Vite, DynamoDB, Docker, Splunk, Claude